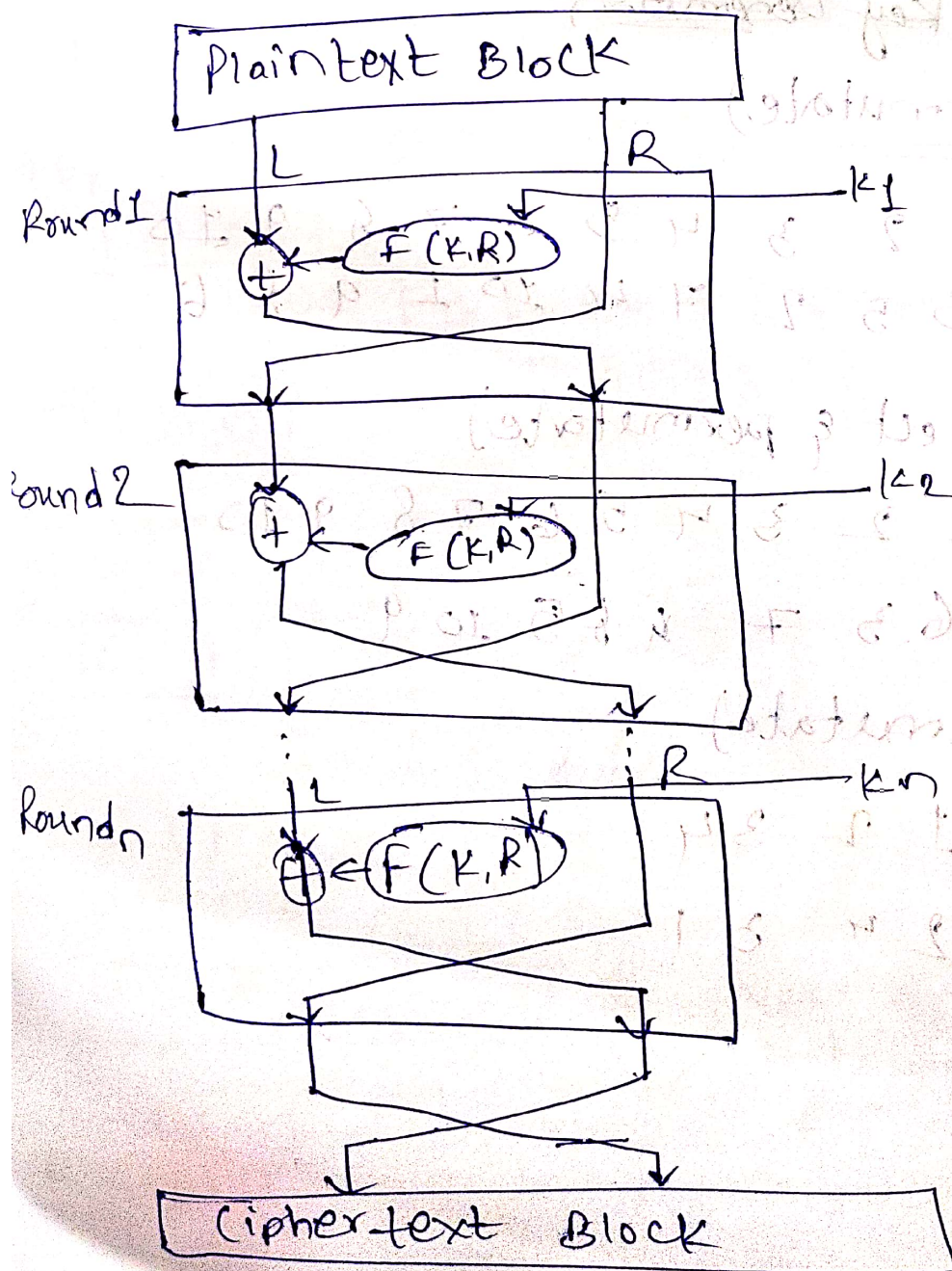
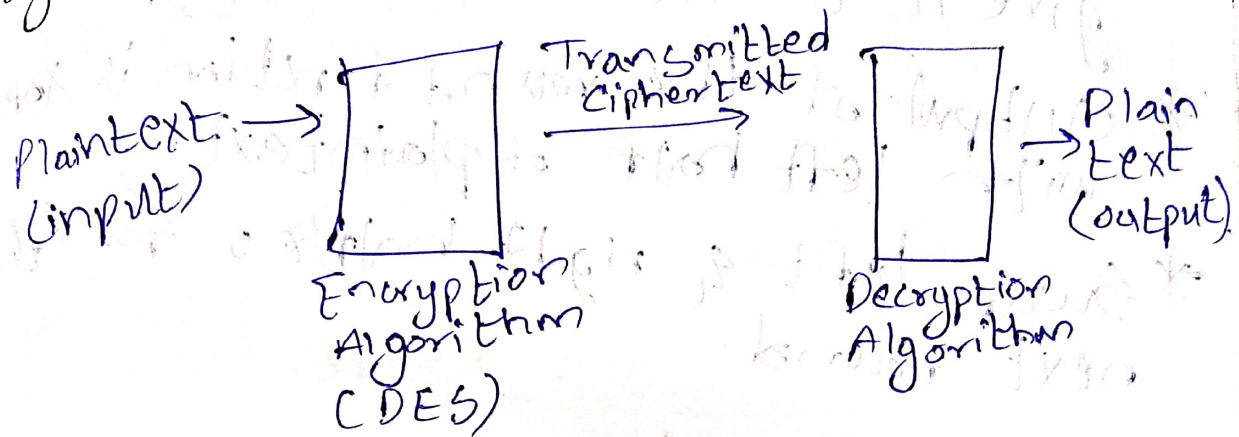


CO-2
SDES (Simplified Data Encryption Standard)
symmetric cipher model



* Plaintext is divided into equal halves left & Right.

* Consider right half of plaintext & give it to round function (F).

* Output of the round function is XORed with left half of plaintext.

* Swap left & right halves for the next round.

DES - Key Generation

P10 (permute)

I/P: - 1 2 3 4 5 6 7 8 9 10
O/P: - 3 5 2 7 4 10 1 9 8 6

P8 (select & permute)

I/P: - 1 2 3 4 5 6 7 8 9 10
O/P: - 6 3 7 4 8 5 10 9

P4 (permute)

I/P: - 1 2 3 4
O/P: - 2 4 3 1

Code obfuscation mini-program

Ex:- P:- 0011 1100

K1: 0011 0110

K2: 1100 1001

IP: 32 14 6 5 8 7

EP: 4 1 2 3 2 3 4

PU: 3 4 1 2

P:- 00 11 11 00
1 2 3 4 5 6 7 8

⇒ 10011100 ⇒ IP applied

⇒ split into two halves,

L = 1001

R = 1100

⇒ Expand & permute
R using EP.

1	1	0	0
1	2	3	4

IP: 4 1 2 3 2 3 4

0 1 1 0 1 0 0 1

↓
XOR with K1.

0 1 1 0 1 0 0 1

0 0 1 1 0 1 1 0

0 1 0 1 1 1 0 1

⇒ Now split this
into two 4-bit
parts.

L = 0101

R = 1111

$$L = 0101$$

$$\text{Row} = 0 \quad L = 1$$

$$\text{Column} = 10 = 2$$

Row 1, column 2 = 1 in 60 matrix
= 01

Now, Repeat the same process with
S1 & right = 1111

$$\text{Row} = 11 = 3$$

$$\text{Column} = 11 = 3$$

$$\text{Row 3, Column 3} = 2$$

$$S0 = 01$$

$$S1 = 11 \Rightarrow S0 + S1 = 0111$$

Apply M4

$$0111$$

$$\begin{array}{r} \cancel{0111} \\ 1234 \end{array}$$

$$3412$$

$$1101$$

$\Rightarrow$ Now XOR this with upper
 $L = 1001$

$$1001$$

$$1101$$

$$\hline 0100$$

*Right half remains unchanged.

$$L'R = 01001100$$

Code obfuscation

maximizing a program's

DES (Data Encryption Standard)

- * Plaintext must be 64 bits in length & key is 56 bits in length.
- 8 bits can be used as parity bits.
- No of rounds = 16.

Process

64-bit Plaintext



Initial Permutation



16 Rounds of DES



Swap Left & Right



Inverse Initial Permutation



64-bit ciphertext.

SAES Algorithm

- 16 bits key is given
- Divided into 4 parts each of 4 bits.
 K_0, K_1, K_2, K_3 .

* 1 nibble = 4 bits

* 1 word = 8 bits

$$K_0 + K_1 = w_0$$

$$K_2 + K_3 = w_1$$

Ex:- P: 0110 1001 1110 0001

K0: 1010 1010 1100 1100

K1: 1111 0000 0000 1111

K2: 1001 0110 0010 1011

Step 1:- $P \oplus K_0$ (XOR)

0110 1001 1110 0001

1010 1010 1100 1100

1100 0011 0010 1101
C 3 2 D

Step 2:- Sub Nibble

1100 $\Rightarrow$ 1100 C

0011 $\Rightarrow$ 1011 B shift 2nd & 4th

0010 $\Rightarrow$ 1010 A

1101 $\Rightarrow$ 1110 E

1100 1110 1010 1011

C E A B

$\begin{bmatrix} C & A \\ E & B \end{bmatrix} \begin{pmatrix} 1 & 4 \\ 4 & 1 \end{pmatrix}$

$(1 \times C) \oplus (4 \times E)$ $(1 \times A) \oplus (4 \times B)$

$(4 \times C) \oplus (1 \times E)$ $(4 \times A) \oplus (1 \times B)$

$\begin{pmatrix} C \oplus D \\ E \oplus F \end{pmatrix} \begin{pmatrix} A \oplus E \\ E \oplus B \end{pmatrix}$

$\begin{pmatrix} 1 \\ 3 \end{pmatrix}$

$\begin{pmatrix} 4 \\ 5 \end{pmatrix}$

0 = 0000

1 = 0001

2 = 0010

3 = 0011

4 = 0100

5 = 0101

6 = 0110

7 = 0111

8 = 1000

9 = 1001

A = 1010

B = 1011

C = 1100

D = 1101

E = 1110

F = 1111



0001 0100 1011 0101
1111 0000 0000 1111

Add Round
Key 1.

1110 0100 1011 0101

E

4

B

A

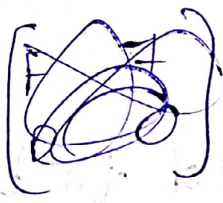
1111 1101 0011 0000

F

D

3

0



F

0

3

D

[F 3
0 D]

1111 0000 0011 1101

1001 0110 0010 1011

Add round
Key 2.

0110 0110 0001 0110



Final ciphertext

Multiple Encryption

* The process of encrypting data more than once using several encryption algorithms or keys is known as multiple encryption.

Ex:- P: 10101010

K1: ~~10101010~~

K2: 00111100

} Double
DES

$P \oplus K_1$

1 0 1 0 1 0 1 0
1 1 1 1 0 0 0 0

$C_1 = 0 1 0 1 1 0 1 0 \oplus K_2$

0 0 1 1 1 1 0 0

$C_2 = 0 1 1 0 0 1 1 0 \Rightarrow$ final ciphertext

Modes of operation

* These are methods used to transform a block cipher into a stream cipher or a hashing algorithm.

→ Electronic Codebook (ECB)

* Plaintext is divided into fixed-size blocks.

* Each block is encrypted independently.

* Ideal for short amount of data.

→ Cipher Block Chaining Mode (CBC)

* Plaintext is divided into fixed-size blocks. An initialization vector (IV) is generated, serving as the initial ~~input~~ input for the encryption process.

* For each plaintext block, the XOR operation is performed between the plaintext block & previous ciphertext.

Cipher Feedback Mode

* In this, the previous cipher text block is used to encrypt the next plaintext block, creating a feedback loop that enables the encryption & decryption.

Output Feedback Mode

* An IV (Initialization Vector) is generated, which serves as the initial input for the encryption process.

Counter Mode

* This is a cryptographic technique used to transform a block cipher into a stream cipher.

* Creates a unique counter value for each block, hence this allows parallel encryption.

Random Number Generators

Criteria for random numbers

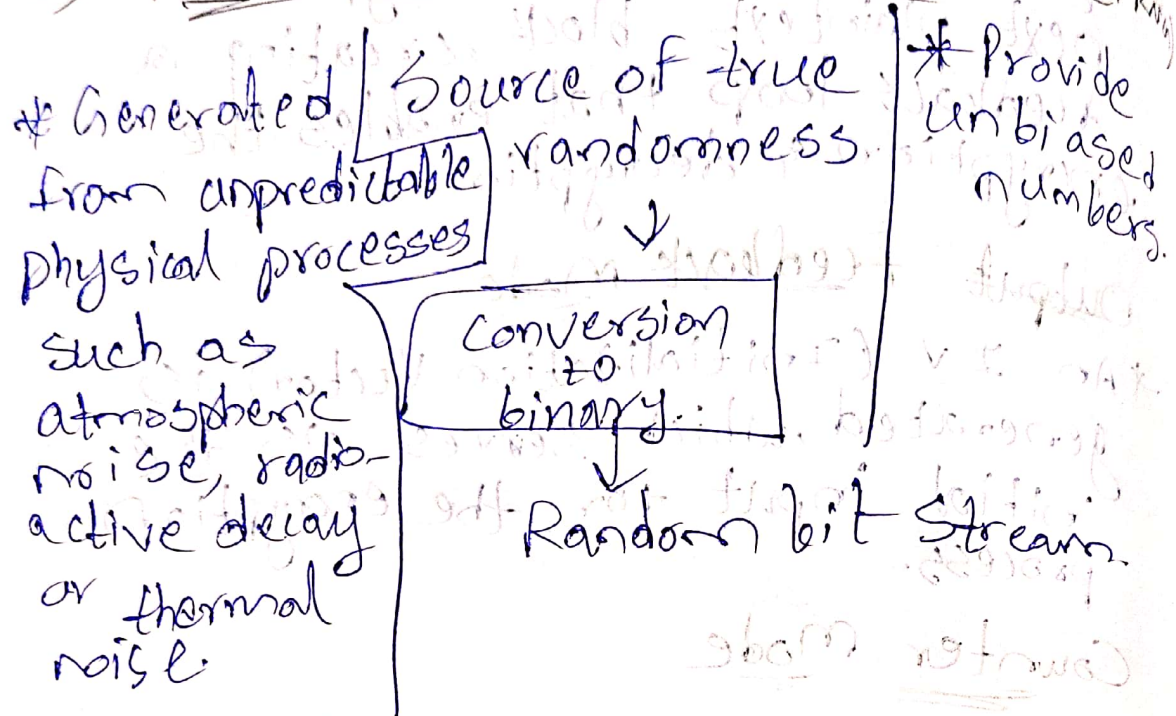
- 1.) Uniform distribution.
- 2.) Independence.

Application of Random Numbers

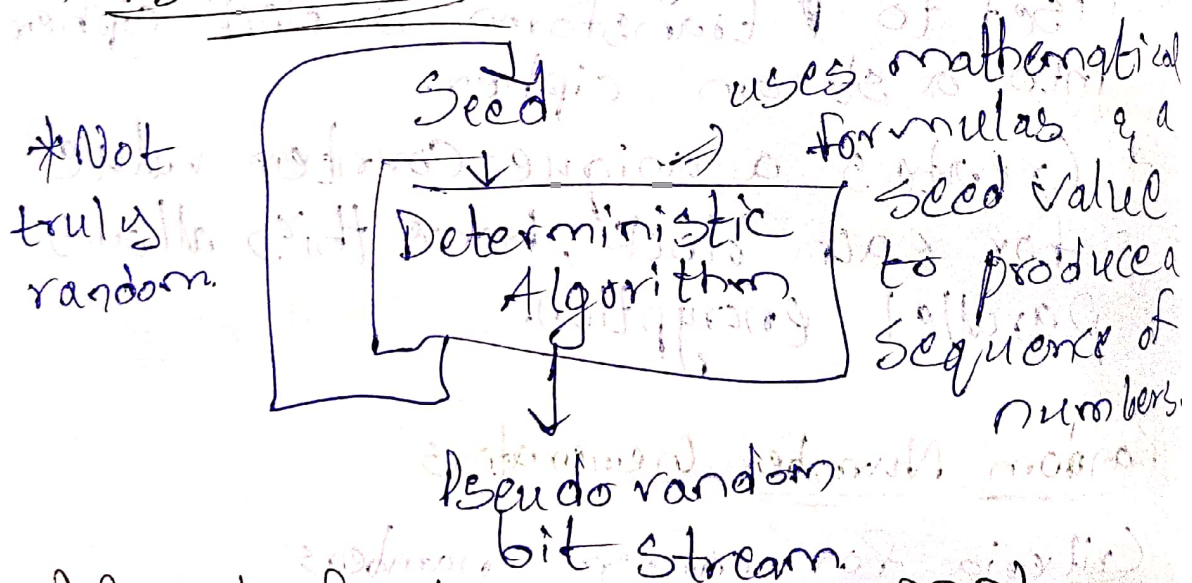
- Generating encryption keys
- Conducting statistical sampling.

Types

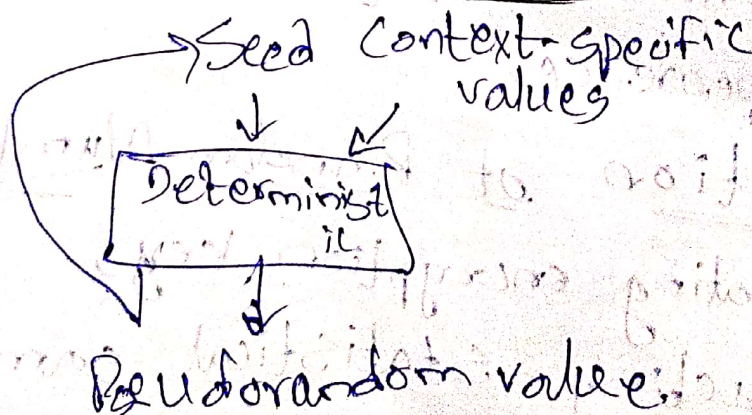
1) True Random Number Generator (TRNG)



2) Pseudo RNG's (PRNG)



3) Pseudo Random Function (PRF)



Types of PRNG's

1) Linear Congruential Generator (LCG)

* Generates random numbers based on a linear equation.

$$X_{n+1} = (a \times X_n + C) \% m$$

X_n = current state.

Ex:- $m=28, a=9, C=2, X_0=8$

$$X_1 = (9 \times X_0 + 2) \% 28$$

$$= (9 \times 8 + 2) \% 28$$

$$= 74 \% 28$$

$$= 18$$

$$X_2 = (9 \times 18 + 2) \% 28$$

$$= 164 \% 28$$

$$= 24$$

$$X_3 = (9 \times 24 + 2) \% 28$$

$$= (216 + 2) \% 28$$

$$= 218 \% 28$$

$$= 22$$

X_{10} :

2) Blum Blum Shub Generator (BBS)

* $X_{k+1} = (X_k)^2 \% n$

$$n = p * q$$

$$X_0 = s^2 \% n$$

Ex:- $P=15, q=8, s=3$

$$n = 15 \times 8 = 120$$

$$X_0 = 3^2 \cdot n$$

$$= 9 \cdot 120$$

$$= 1080$$

$$X_{\cancel{1}} = (X_0)^2 \cdot n$$

$$= 81 \cdot 120$$

$$X_1 = 81$$

$$X_2 = 81^2 \cdot 120$$

$$\vdots = -$$

Stream Ciphers

- * Type of symmetric encryption algorithm that operates on individual bits or bytes of data.
- * Encrypt data on continuous stream.
- * Used for securing network communication.

SRC4

- * Referred to as Rivest Cipher 4. is a systematic stream cipher.
- * Two 8-byte Arrays are taken S & T from 0 to 7.